Car Minus Driver: Autonomous Vehicles Driving Regulation, Liability, and Policy

By Damien A. Riehl

"Take me to court." Today, a lawyer might say that to a contentious adversary (hopefully not), to a ride-sharing driver (maybe), or to a phone for directions (most likely). Tomorrow, lawyers may well direct similar commands to their self-driving vehicles.

Automated driving is coming—and in some respects, it's already here. Our roads already have vehicles with automated-driving technologies, ranging from electronic stability control (mandatory since 2011) to Tesla's "Autopilot," which uses "autosteer" to "keep[] the car in the current lane," to "parallel park itself," and to engage "traffic-aware cruise control" to "intelligently determine[s] the vehicle's behavior in that moment relative to its surroundings."[1] Authorities have reached a general consensus that automation-assisted driving will continue along this spectrum—from assist to control—until vehicles are capable of full autonomy. Manufacturers have promised to bring millions of automated-driving vehicles to our roads by 2020.[2]

Americans love their cars. So during this time of transition, lawyers, judges, lawmakers, and policymakers must decide how they will react to this seismic societal shift. Should our approach be *laissez faire*, allowing development through common law? Or should the law be shaped proactively through regulation or legislation? To best resolve these questions, decisionmakers should first know the technology. As Judge Easterbrook has noted about lawyers and new technologies: "The blind are not good trailblazers."[3]

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Automated Driving: An Evolution of the Internet of Things

Society is awash in the Internet of Things—smart, connected devices permitting us to perform tasks previously unimaginable—and few developments demonstrate the power of the Internet of Things more dramatically than automated driving.[4] The US Department of Transportation (DOT) and National Highway Traffic Safety Administration (NHTSA) have stated that automated driving "may prove to be the greatest personal transportation revolution since the popularization of the personal automobile nearly a century ago."[5]

In the 1950s, futurists predicted flying cars, but the concept of fully autonomous vehicles was still beyond the scope of most imaginations until the 2000s. In 2007, six teams competed in the Defense Advanced Research Projects Agency (DARPA) Urban Challenge to push the limits of vehicle autonomy, and around 2009, Google and other companies started moonshot initiatives to develop self-driving vehicles. It's easy to imagine this technology giving new power to the blind and disabled, to the elderly, and to children who may never need drivers' licenses.

As with many new technologies, the law must catch up with the issues raised by this brave new world: Crashes today are usually caused by people, and tomorrow they might be caused by algorithms. But while algorithms might cause some crashes, automated-driving proponents claim that the systems will still be far safer than humans—potentially cutting driving deaths in half and saving an estimated 16,000 lives annually. How society responds to these developments will shape many legal arenas, including regulations, insurance, statutes, and the common law.

The road to our driverless future is currently being paved. To paraphrase author William Gibson: Automated driving is already here—it's just not evenly distributed. The DOT and NHTSA have stated that automated driving's "rapid development" has included "partially and fully automated vehicles . . . nearing the point at which widespread deployment is feasible."[6] This revolution includes the largest automobile manufacturers (*e.g.*, Audi, BMW, Ford, GM, Honda, Mercedes, Nissan, Tesla, Toyota, Volkswagen, and Volvo) and the largest software companies (*e.g.*, Google, Apple). The promise of a multi-billion-dollar market has spawned a 21st-century space race: Tesla hired

a prominent Apple software architect to lead its Autopilot Engineering division,[7] GM invested $500 million in a joint venture with ride-sharing service Lyft to develop self-driving cars.[8] And Uber has partnered with Carnegie Mellon University on the company's mission to end car ownership and help create our "driverless future."[9]

Some automated-driving advancements have already made their way onto public roads. In October 2015, Tesla began allowing owners (not just test drivers) to enable "Autopilot" mode,[10] which is not fully autonomous but seeks to assist drivers, whom Tesla encourages to keep their hands near the wheel and eyes on the road.[11] GM announced that its similar "super cruise" system will be deployed in 2017.[12] Toyota has introduced a suite of automated systems, including pedestrian detection and "pre-collision systems" that can apply brakes automatically.[13] Automated cars from Google and Waymo (subsidiary to Google's parent company, Alphabet) have driven more than 4 million miles on public roads—2 million of that the past year.[14] Surveys have shown that US drivers increasingly embrace the concept of riding in automated-driving vehicles.[15]

Automated driving's early days have not been without speed bumps. February 2016 saw Google/Waymo's first automation-caused crash, a fender-bender with a bus.[16] In May 2016, a driver using Tesla's Autopilot mode became the first automated-driving fatality in over 130 million miles.[17] (In comparison, standard US vehicles have one fatality every 94 million miles, and worldwide it's every 60 million miles.)[18] In June 2017, the National Transportation and Safety Board (NTSB) concluded that the Tesla accident was caused by a combination of both the technology and the driver.[19]

Despite these potential setbacks, the federal NHTSA has been undeterred, stating that automated driving can save thousands of lives: "No one incident will derail the DOT and NHTSA from [their] mission to improve safety on roads through new life-saving technologies."[20]

The Self-Driving Future Will Arrive Gradually

Our conversion to automated driving will not occur overnight, but the transition will come sooner than many imagine, and progress will be steady. The shift will likely follow the path of prior advancements such as the Internet and smartphones: What begins as luxury quickly becomes necessity. Gradually, assistive technologies (*e.g.*, adaptive cruise control, crash avoidance) may shift to full autonomy. Or manufacturers might instead take Google/Waymo's approach, choosing to jump straight to full autonomy.

Adoption predictions are ambitious. One report estimates that by 2020, automated vehicles will number 10 million.[21] Most engineer-respondents to studies by the Institute of Electrical and Electronics Engineers (IEEE) and Information Handling Services (IHS)[22] believe that manufacturers will remove rear-view mirrors, horns, and emergency brakes by 2030—and steering wheels and gas/brake pedals will be gone by 2035.[23] Some estimate that our fully driverless future will arrive around 2050.[24]

In the near term, the automated-driving transition will likely be gradual, moving along a spectrum from fully manual to partial assistance to fully automated. This gradation was standardized in 2013, when the international engineering vehicle-standards organization Society of Automotive Engineers (SAE) International[25] defined the levels of driving automation.[26] In September 2016, NHTSA—which had in 2013 created its own distinct definitions, using only four levels—adopted the SAE's levels and definitions. By September 2017, federal regulators had evolved the automation-level definitions to the following:

> L0 **No automation.** Zero autonomy; the driver performs all driving tasks.
>
> L1 **Driver assistance.** Vehicle is controlled by the driver, but some driving assist features (*e.g.*, steering, speed) may be included in the vehicle design.
>
> L2 **Partial automation.** Vehicle has combined automated functions (*e.g.,* acceleration and steering), but the driver must remain engaged with the driving task and monitor the environment at all times.
>
> L3 **Conditional automation.** Driver is a necessity, but is not required to monitor the environment. The driver must be ready to take control of the vehicle at all times with notice.
>
> L4 **High automation.** The vehicle is capable of performing all driving functions under certain conditions. The driver may have the option to control the vehicle.
>
> L5 **Full automation.** The vehicle is capable of performing all driving functions under all conditions. The driver may have the option to control the vehicle.[27]

Google/Waymo and Tesla are striving, even at this early stage, to jump toward full autonomy. In November 2017, Waymo deployed Level 4 vehicles on Arizona public roads without a "safety driver."[28] Tesla's founder Elon Musk has claimed that soon, owners will be able to "summon" their cars to arrive by driverless cross-

country trip (*e.g.*, New York to LA).[29] Google/Waymo has also designed its fleet for full autonomy (removing conventional steering wheels, accelerators, brakes, etc.), expressing concern that permitting human intervention might hinder (not help) safety:

> Google[] belie[ves] that the SDS [self-driving system] consistently will make the optimal decisions for the SDV [Self-Driving Vehicle] occupants' safety (as well as for pedestrians and other road users), the company expresses ***concern*** that ***providing human*** occupants of the vehicle with ***mechanisms*** to ***control*** things like steering, acceleration, braking, or turn signals, or providing human occupants with information about vehicle operation controlled entirely by the SDS, could be ***detrimental to safety*** because the human occupants could attempt to override the SDS's decisions.[30]

Google/Waymo supported this conclusion by pointing to its automated-driving fleet's 1.3 million miles driven between 2009 and January 2016, during which all 17 crashes were caused by human error.

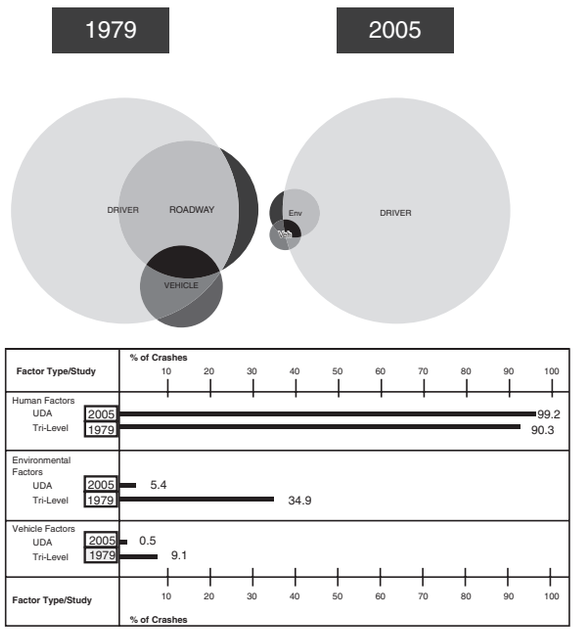
While consumer adoption could be swift, the shift to automated driving will likely be even faster for commercial driving—starting with ride-sharing, taxis, public transportation, and long-haul trucking.[31] For example, in February 2016 a convoy of self-driving trucks made its way across Europe,[32] signaling a potential shift in land-based shipments. In November 2017, Tesla announced that Tesla Semi could "platoon," where the leading truck has a human driver—but trailing trucks would be driverless, enjoying reduced wind resistance and cost.[33] Labor currently constitutes about 75 percent of trucking costs, and regulations require human drivers to take 8–hour breaks every 11 hours. So 24–hour non-human driving could cost 75 percent less for double the productivity (a 400 percent price-performance improvement).[34] Similar market factors will likely affect other areas of commercial driving: taxis, buses, and other public transportation.

Potential and Economic and Safety Effects

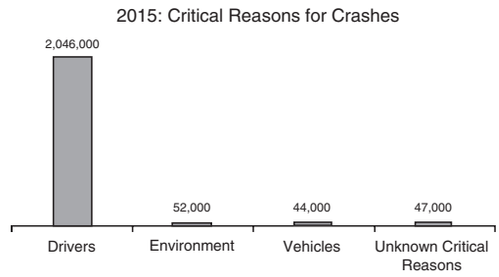
It's no secret that driving has always been unsafe. Those old enough to remember Ralph Nader's book *Unsafe at Any Speed* have seen giant leaps in vehicle-safety improvements. But even today, US auto crashes still kill about 32,000 people per year (88/day), or nearly three times the number who die in firearm homicides.[35] For ages 5 to 34, crashes are the leading cause of death.[36]

Automakers have improved vehicle safety, but not its weakest link: humans. Federal studies reflect that over 90 percent of crashes are caused, at least in part, by human error. A 1979 federal study found that "human errors and deficiencies" caused 90 to 93 percent of investigated crashes;[37] a 2001 federal study found that "a driver behavioral error caused or contributed to" 99 percent of the crashes investigated;[38] and a 2005 study found that in causing crashes, vehicle and environmental factors were dwarfed by "human factors." As cars become safer (*e.g.*, anti-lock brakes, electronic stability control, side airbags, adaptive cruise control, structural design), the percentage of crashes caused by faulty vehicles and environmental factors shrink, and the percentage of human-caused crashes increases commensurately. The graph and table shown in Exhibit 1 compare the 1979 study (Tri-Level) and 2005 study (UDA) on overlapping crash causes, on a percentage basis.

Exhibit 1 Comparison of Overlapping Crash Causes



2015 NHTSA Study:



Even in its infancy, automated driving has been surprisingly safe. Some studies support the argument that automated driving is currently safer than human driving[39] (*e.g.*, in 2017, Elon Musk claimed that Tesla's systems were "10 times safer than a human driver"), while others disagree[40]—and most studies acknowledge that available data is limited (*e.g.*, unreported human-driven crashes, automated driving's comparatively low miles driven). But one thing remains certain: The technology underpinning automated driving will improve, likely exponentially.

Proponents estimate that automated driving could prevent 90 percent of traffic crashes, improve commute times, and conserve energy—annually preventing 5 million crashes and 2 million injuries, reducing collective commute times by 4.8 billion hours, saving 1.9 billion gallons of fuel, and drastically reducing the currently estimated $300–$450 billion in crash-related costs and lost productivity.[41]

Benefits would likely even be realized in the earliest days of automated-driving technology, before its perfection. A 2017 RAND study ran three computer-modeling scenarios, and automated driving improved on human driving by 10 percent, 75 percent, and 90 percent. Even the 10 percent improvement would save tens of thousands to hundreds of thousands of lives.[42] That study bolsters the federal government's approach: "Waiting for perfect" appears to be less desirable than "full speed ahead."

Computers do not share human drivers' foibles: They cannot be inebriated, they don't text, and they don't fall asleep. Automated-driving systems can also have super-human qualities: 360–degree vision; 100 percent alert time; constant communication with the road, traffic lights, and other cars; "sight" through fog and darkness; and universal, system-wide routing for traffic-flow optimization. Computers react faster: Humans' reaction time is approximately 1.5 seconds,[43] while computers' reaction times are measured in milliseconds (and, per Moore's Law, improving exponentially).

Of course, manufacturers and software designers also have good reason to reduce the risk of crashes. Liability concerns incentivize designers to err toward caution, slower speeds, and preventive stops. For example, in 2015 an officer stopped a Google/Waymo automated vehicle for driving 10 mph below the speed limit.[44] A popular video purports to show an autopiloted Tesla—traveling at 45 mph on a rainy night—automatically stopping to avoid a vehicle that veered in its path.[45]

In addition, automated-driving systems learn collectively. Human drivers draw from individual experience, often making the same mistake as thousands made earlier. But automated vehicles can automatically incorporate data from errors and driving conditions from millions of miles driven by *other* automated vehicles. NHTSA has noted that whenever an automated vehicle encounters an "edge case" for which it was unprepared, "that data can be taken, analyzed, and then the lessons can be shared with more than the rest of that vehicle fleet" and "all automated vehicles."[46] As such, automated-driving systems could improve the problem of the 16–year-old human driver making the same mistake made by thousands before. Instead, automated driving can harness the exponential power of crowdsourcing, big data, and Moore's law to collectively perfect auto safety.

Potential Brakes and Speed Bumps

The automotive and insurance industries have identified six possible areas that could slow the advancement of automated driving. The first three obstacles are the most significant: (1) legal liability, (2) policymakers, (3) consumer acceptance, (4) cost, (5) infrastructure, and (6) technology. Of course, the first two are the primary (if not exclusive) province of lawyers, judges, and government officials. As such, our profession should consider the state of the technology, determining which policies and caselaw standards (if any) should shift to address this fast-moving development.

Regulations and Statutes

Of course, even the safest automated-driving systems will have crashes, caused by either machines or humans, and where there are damages, questions of liability quickly follow. As such, regulation is inevitable. But one scholar cautions against requiring even more of automated systems than our laws currently require of human drivers: "I'm concerned about computer drivers, but I'm terrified about human drivers."[47]

Society's appetite for automated driving (or lack thereof) will largely determine how quickly we will achieve automated driving's promised potential safety benefits. A RAND study indicates that to achieve automated driving's potential effects (*e.g.,* increased manufacturer liability, decreased personal liability) and benefits (*e.g.*, improved traffic flow, lower driver error, increased energy efficiency, and fewer vehicle crashes), proponents must overcome certain risks.[48] Topping the list are (1) inconsistent state regulations and (2) liability issues.[49]

To Regulate or Not to Regulate?

The question of which entities will primarily regulate automated driving is currently unanswered: Will the applicable statutes and regulations be driven mostly by

states, by the federal government, or by a combination of the two?

Currently, auto insurance is largely self-regulated, and each jurisdiction enacts its own statutes and regulations (roughly split between tort states and no-fault states). But that element of "fault" is premised on the primary *cause* of fault: humans. Because automated-driving systems will shift that analysis toward manufacturer liability, and because vehicles are products used across state lines, automated-driving regulations may well shift to the federal level. Manufacturers also may welcome federal regulation, which could avoid the costly necessity of complying with the laws of 51 jurisdictions.[50] But the plaintiffs' bar may well encourage state regulation, permitting injured clients to benefit from a more-favorable state's laws.

Do Automated-Driving Systems Require Enabling Legislation?

While some states have enacted automated-driving legislation, an untested question is whether automated driving is legally permitted even *without* legislation. A *New York Times* op-ed piece in 2011 argued a common perception: "The driverless car is illegal in all 50 states."[51] But legal scholar and automated-driving expert Professor Bryant Walker Smith opines that absent regulation, automated driving may well be permitted under the legal principle that "everything which is not forbidden is allowed."[52] Professor Smith argues that three legal regimes (the 1949 Geneva Convention on Road Traffic, NHTSA regulations, and all states' vehicle codes) lead to the conclusion that automated driving is likely permitted, even today.[53]

1949 Geneva Convention on Road Traffic

The US adheres to the 1949 Geneva Convention on Road Traffic, which requires signatory countries to enforce certain tenets, including Article 8, which requires that "drivers shall at all times be able to control their vehicles."[54] This standard is likely satisfied if the system permits human intervention, as under Levels 1–4. But fully autonomous Level 5 may well run afoul of the Geneva provision, unless manufacturers enable human override. An open question: Are fully autonomous Level 5 systems properly considered "drivers"—especially if they perform as well as (or better than) humans?

DOT/NHTSA Regulation

Beyond the Geneva Convention, federal regulators have encouraged automated driving's expansion. The DOT and NHTSA have released successive guidance on automated vehicles: The 2013 Preliminary Statement of Policy Concerning Automated Vehicles (2013 NHTSA Statement),[55] updated in January 2016 by its Policy Statement of Policy Concerning Automated Vehicles, its Federal Automated Vehicles Policy (2016 FAVP), and its Automated Driving Systems 2.0 (2017 ADS).[56] These regulatory statements and guidance (but not formal rulemaking) are the primary indicators of NHTSA's nascent automated-driving policy, which the agency has said will be nimble and responsive.

Even absent formal automated-driving rulemaking, the federal regulators today permit manufacturers to demonstrate compliance with the Federal Motor Vehicle Safety Standards (FMVSS) through self-certification.[57] This has been the process for prior safety advancements (*e.g.*, airbags), but rapidly evolving automated-driving technology can make even basic assumptions quickly obsolete, requiring regulatory revision.

In February 2014, the federal agencies approved related vehicle-to-vehicle (V2V) communications systems, where vehicles sense each other's speed and location to help avoid crashes.[58] Similar benefits come from vehicle-to-infrastructure (V2I) sensors, which analyze smart roadways, traffic lights, and other infrastructure. The DOT estimates that V2V will prevent 76 percent of crashes.[59] A provision of 2015's Fixing America's Surface Transportation (FAST) Act[60] established a Federal Highway Administration fund to accelerate deployment of automated-driving technologies.[61] Legislators are also considering privacy: The Autonomous Vehicle Privacy Protection Act of 2015 would have studied DOT readiness to address the challenges accompanying automated driving, including those to consumer privacy protections. But that bill failed.[62]

In February 2016, NHTSA responded to a letter from Google requesting interpretation of several of the 2013 NHTSA Policy's provisions.[63] One issue, for example, involved a current regulatory requirement that vehicles have a foot-controlled brake, which Google's fully autonomous design lacks (because Google believes that human controls would actually hinder, not help, safety).[64] NHTSA responded that Google's design would not comply with the existing regulation's plain language; that in light of "changed circumstances," the agency will consider rulemaking to modify the regulations; and that "Google may wish to consider petitioning the agency for an exemption from these provisions."[65] Industry innovates, and regulators respond.

In regulating the automated-driving space, NHTSA has said it will move quickly. Responding to public suggestions that the agency may be moving too rapidly, NHTSA has consistently noted its traditional approach: Allow automakers to introduce new safety improvements (*e.g.*, airbags, electronic stability control, cruise

control) quickly, and then after some time, NHTSA implements new safety standards.[66] But where manufacturers' implementation of earlier safety technology (*e.g.*, airbags) took years, improving automated-driving software through over-the-air (OTA) updates can take mere minutes.[67] NHTSA has concluded that because automated driving has the potential to save many lives, "wait[ing] for perfect" would mean "waiting for a very, very long time," asking "How many lives might we be losing while we wait?"[68] Through a "nimble and flexible" approach, NHTSA seeks to "keep pace with technological innovation" and "provide certainty to manufacturers and developers."[69]

In September 2016, NHTSA released its Federal Automated Vehicles Policy (2016 FAVP), which served as "guidance rather than . . . rulemaking."[70] The agency also promised to "speed the delivery of an initial regulatory framework." That 2016 FAVP guidance sought to provide flexible measures, giving the industry some latitude to deploy automated-driving systems. The policy included several elements:

- **Vehicle performance guidance**, including a 15-point "Safety Assessment" for design, development, testing, and deployment;
- **Model state policies**, including recommendations to states for legislation;
- **Clarified existing rules** and how they apply to automated driving;
- **New rules and authorities** that NHTSA may consider seeking in the future.

In the 2016 FAVP, the federal agencies committed to streamline their review process by issuing periodic "simple" regulatory interpretations and exemption requests. NHTSA stated that it would use feedback and new data to update the policy "within the next year."

In September 2017 and after an administration change (from Obama to Trump), NHTSA did replace the 2016 FAVP by releasing updated guidance entitled, "Automated Driving Systems 2.0" (2017 ADS). The updated guidance did not completely overhaul the 2016 FAVP; rather, the newer guidance largely provided lessons learned over the past year. Notably, as shown in Exhibit 2, the Obama administration's 15-point "Safety Assessment" was changed by the Trump administration to 12 "Safety Design Elements"—eliminating the points Privacy, Registration and Certification, and Ethical Considerations.

Exhibit 2—Obama's 15-Point Safety Assessment vs. Trumps' 12 Safety Design Elements

2016—15-point "Safety Assessment"	2017—12 "Safety Design Elements"
• Data Recording and Sharing	• Data Recording
• System Safety	• System Safety
• Vehicle Cybersecurity	• Vehicle Cybersecurity
• Human Machine Interface	• Human Machine Interface
• Crashworthiness	• Crashworthiness
• Consumer Education and Training	• Consumer Education and Training
• Post-Crash Behavior	• Post-Crash ADS Behavior
• Federal, State and Local Laws	• Federal, State, and Local Laws
• Operational Design Domain	• Operational Design Domain
• Object and Event Detection and Response	• Object and Event Detection and Response
• Fall Back (Minimal Risk Condition)	• Fallback (Minimal Risk Condition)
• Validation Methods	• Validation Methods
• Privacy	~~• Privacy~~
• Registration and Certification	~~• Registration and Certification~~
• Ethical Considerations	~~• Ethical Considerations~~

The updated 2017 ADS emphasized that as autonomous vehicles are deployed, the tenets of federalism should favor the federal government, not states:

> NHTSA strongly encourages States not to codify this Voluntary Guidance (that is, incorporate it into State statutes) as a legal requirement for any phases of development, testing, or deployment of ADSs [Automated Driving Systems]. Allowing NHTSA alone to regulate the safety design and performance aspects of ADS technology will help avoid conflicting Federal and State laws and regulations that could impede deployment.

The extent to which states will comply with the federal regulator's admonishment remains to be seen.

Importantly, the 2017 ADS guidance continues to be voluntary, not mandatory. It notes that although manufacturers are "encouraged" to submit assessments, they are "not subject to Federal approval."[71] Even though automated-driving vehicles have operated in many states for years, by September 2017, federal regulators apparently had not yet received a single Safety Assessment.[72] After 2016 FAVP's suggested "assessment" was downgraded to a mere list of "design elements," how many manufacturers will adopt that voluntary expense?

Federal Legislation

Nearly all the federal work in automated driving has been administrative, but Congress has recently made overtures to creating federal legislation to guide the regulatory agencies. In September 2017 (within days of NHTSA's 2017 ADS guidance), the US House of

Representatives passed a bill called the SELF DRIVE Act. A few weeks later, the Senate unveiled similar legislation entitled the American Vision for Safer Transportation Through Advancement of Revolutionary Technologies (AV START) Act. The two bills were similar, but they contained significant differences. By early 2018, the Senate version had not yet been passed.[73]

State Regulations

While the federal government continues its regulatory evolution, some individual jurisdictions have passed statutes and regulations to define which automated-driving activities are permitted and which are prohibited. By early 2018, the number of states passing legislation related to autonomous vehicles numbered 21.[74] In 2011, Nevada became the first state to expressly permit automated driving. Since then, many states have considered similar legislation, but few bills have passed. Stanford University's Center for Internet and Society tracks state legislative and regulatory action surrounding automated driving,[75] and its map is provided in Exhibit 3.

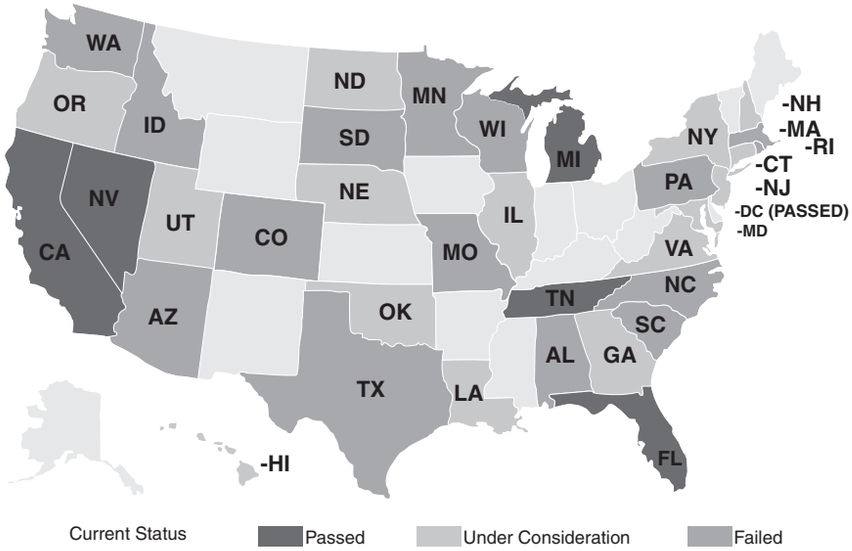
Most states have either enacted or considered laws on automated driving. Several jurisdictions (including California, District of Columbia, Florida, Michigan, Nevada, North Dakota, Tennessee, and Utah) have passed legislation. Broadly, the existing state regulations define autonomous driving, sometimes expressly permitting it and sometimes authorizing studies. The states that have expressly regulated automated driving have permitted it (*e.g.*, California's statement that it "presently does not prohibit or specifically regulate the operation of autonomous vehicles"), and to date, no states have barred automation.

Manufacturers seeking to comply with a 51–jurisdiction patchwork may have some difficulty, given the potential to run afoul of the most-restrictive jurisdictions. For example, a New York driving statute[76] (enacted in the 1970s) requires a driver to keep "at least one hand" on the steering wheel.[77] (A 2016 bill sought to update the law to permit "driving technology . . . to perform the steering function," but the bill died in committee.)[78] And the District of Columbia prohibits distracted driving and mandates a driver's "full time and attention."[79] Fully autonomous Level 5 vehicles—such as Google's prototypes, which currently lack traditional steering wheels and are indifferent to driver attention—may not comply with those existing statutes' plain language.

Federal policymakers have encouraged states to establish "a consistent national framework rather than a patchwork of incompatible laws." NHTSA and DOT seek to avoid a future where manufacturers would need to develop "50 different versions" of automated vehicles. To that end, federal regulators have provided states with a Model State Policy intended to guide state lawmakers.[80]

A 50-state survey of automated-vehicle laws would exceed this article's scope, particularly when Web sites such as Stanford's Center for Internet and Society perform the hard work of tracking and updating related

Exhibit 3—State Legislation on Automated Driving



legislation and regulation.[81] As such, this article summarizes some Midwest states' approaches to automated driving regulation, discussed below.

Minnesota: Failed Bill to Study Automated Driving's Potential

The Minnesota legislature has introduced several bills related to automated driving, but all died in committee. In the 2013 session, the first such bill, HF 1580, sought to "Direct[] the commissioner of transportation to 'evaluate policies and develop a proposal for legislation governing regulation of autonomous vehicles' by January 31, 2014."[82] But that bill died after its introduction and first reading.[83]

The second attempt came in the 2016 session, when companion bills, SF 2569 and HF 3325, sought to create and support an "autonomous vehicles task force" to "serve mobility needs of people with disabilities."[84] In April 2016, committees in both houses recommend passage as amended, but the bills both died before a vote.[85]

In March 2018, Minnesota Governor Mark Dayton issued an executive order relating to automated and connected vehicles.[86] Executive Order 18-04 establishes an advisory council that will study, assess, and prepare for automated vehicles—ultimately developing "recommendations for changes in state law, rules, and policies to maximize the benefits and prepare for the widespread adoption of automated and connected vehicles."[87] The resultant report is scheduled to be complete in December 2018.

If past attempts are any indication, the Minnesota legislature appears willing to consider *studying* automated driving through administrative policy, proposal, or task force, but Minnesota lawmakers do not appear to have an appetite to quickly regulate the issue.

North Dakota: Funded Studies on Automated Driving for Potential Future Legislation

In March 2016, the North Dakota legislature became Minnesota's first neighboring state to legislatively act on automated driving, by passing "an act to provide for a legislative management study of automated motor vehicles."[88] The law asks legislative management to "consider studying what, if any, current laws need to be changed to accommodate the introduction or testing of automated motor vehicles."[89] The direction relates to full automation (Level 5), and it also seeks to study potential effects on safety and wellbeing: "reduc[ing] traffic fatalities and crashes," "reducing or eliminating driver error," "reduc[ing] congestion," and "improv[ing] fuel economy."[90] The bill seeks recommendations and any draft enabling legislation in time for the legislature's 2018 session.[91]

In 2017, North Dakota's legislature passed another bill, HB 1202, which directs the North Dakota Department of Transportation to collaborate with the "autonomous vehicle technology industry" to conduct another study and deliver another report—along with draft legislation—for the legislature's 2018 session.[92] The study and report will relate to highway use of automated vehicles, as well as reviewing how that use might interact with current laws and regulations.[93]

Wisconsin: Failed Bill but Signed Executive Order

In March 2013, Wisconsin legislators introduced a bill (SB 80) to create rules explicitly authorizing "the operation of autonomous vehicles on the highways" if they met particular requirements.[94] For example, the bill would have required human presence and intervention, a human driver's license, $5 million liability-insurance coverage, and other technical requirements.[95] The bill failed to pass the Senate.[96]

In May 2017, Wisconsin Governor Scott Walker signed Executive Order #245, creating a committee to advise the governor "on how best to advance the testing and operation of autonomous and connected vehicles in the State of Wisconsin."[97] Areas for the committee to examine include: "vehicle registration, licensing, insurance, traffic regulations, equipment standards, and vehicle owner or operator responsibilities and liabilities under current law."[98]

South Dakota: Failed Bill to Authorize Manufacturers' Testing of Automated Driving

In 2014, South Dakota legislators considered a bill to "authorize the testing of autonomous cars on the [state's] highways."[99] Senate Bill 139 would have permitted manufacturers to test automated vehicles in the state.[100] After three weeks, the bill was tabled and died in committee.[101]

Hybrid: Broad Federal Regulations, State-Specific Judicial Gap-Filling

Some commentators have suggested that the best approach may be for NHTSA to create a comprehensive federal regulatory regime (prescribing basic safety criteria) and to permit states to fill any gaps in the regulatory regime through specific statutes and judicial decisions.[102] That method would allow manufacturers to deploy potentially life-saving technology more quickly, while still permitting state legislatures to analyze locally specific use cases and the judiciary to develop the law for less-common edge cases.

Of course, legislation and regulation are only two factors in the legal equation. Courts and litigators must also determine whether and how automated driving will affect liability. As computer algorithms decide vehicle actions, will auto-crash cases shift from pure human neg-

ligence to product-liability analyses (such as manufacturer negligence, manufacturing defects, design defects, failure to warn, breach of warranty, and misrepresentation)? Will any liability be limited by end-user license agreements (EULAs)? How will insurers adapt to automated driving's promised safety benefits? How will law enforcement adjust to automated vehicles' driving-law adherence, which is likely to limit the use of pretextual stops? If automated vehicles log all routes, how will that affect privacy? When manufacturers' algorithms—created months and years in advance—can make driving decisions that will essentially choose who will live and who will die in certain circumstances, how should manufacturers make those algorithm-creation choices? Those topics and more are discussed below.

Potential Liability: Shift from Human Negligence to Product Liability

The federal government's apparent appetite to permit the technology's quick deployment, before exhaustive regulations could even be crafted—and because state regulations are either non-uniform or non-existent—litigators and courts will likely play a large role in shaping the common law's approach to automated driving. This section discusses potential liability shifts, as well as automated driving's potential effects on insurance, criminal law, privacy law, and ethics.

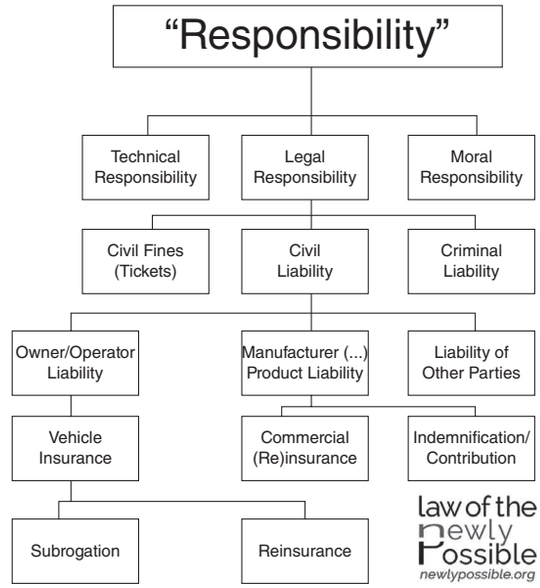
Automated-driving crashes are inevitable, so courts will likely need to determine the legal and factual standards for determining liability. Depending on the automation level, the legal and factual analyses will likely differ substantially from today's legal analyses.

Fully manual Level 0 is the status today, and Level 5 will be our autos' self-driving future. But in the interim, much of the litigation will focus on Levels 1–4. As long as humans and computers are co-pilots, determining "cause" will become increasingly complex.

Even in these early days, we see human-machine finger pointing playing out. In January 2018, a driver arrested for drunk driving apparently explained that his "Tesla had been set on autopilot."[103] Tesla responded that its autopilot function should be used "only with a fully attentive driver."[104] These claims will continue.

During this time of transitional human-machine cooperation, insurance and litigation will likely develop criteria for determining a crash's cause(s). Did the human override the computer? Was the human paying attention—and to a sufficient degree? Did the computer properly alert the human? Did the computer notify the human early enough? Did the computer misinterpret data? The fact-specific queries and aggregate test cases to determine liability will likely be crafted by insurers, litigators, and regulators.

Exhibit 4—Potential Responsibility for Automated Vehicle Accidents



Today, one factor that insurers and litigators consider when determining potential liability is a vehicle's make and model (*i.e.,* "hardware"). As the automated systems assume more control, an increasingly important factor will be the automated vehicles' software—in terms of both the version's components and the updates' timeliness. Did the software company sufficiently update the algorithms to reflect evolving best practices? If so, did the manufacturer push the algorithm updates to all vehicles? If so, did the vehicle owner *accept* the update? Should the onus be on the owner to update the algorithms, or should the manufacturers design the systems to push updates automatically? (Modern software, operating systems, and browsers increasingly push updates without user intervention, ensuring that users integrate the most-recent versions, and their security updates, automatically.)

One insurance-industry group contemplates a future where burdens of proof might shift from drivers to manufacturers, who may be required to prove that their automation did ***not*** cause a crash. As shown in Exhibit 4, the Web site "Law of the Newly Possible" provides a helpful graph mapping potential responsibility for automated-vehicle crashes, noting that a significant actor in the civil-liability analysis will be insurance companies.

Steering Toward Liable Parties

As with any emerging technology, the list of potential defendants may be initially broad. But as the law develops, liability may coalesce to certain defendant categories. Professor Bryant Walker Smith assesses that in

a single crash, potential defendants might include vehicle owners, drivers, manufacturers, sensor suppliers, and data providers.[105]

Vehicle Owners: Override, Failure to Maintain

Of course, a vehicle owner might bear liability in several contexts. In Levels 1–4, owners might override the automation negligently, purposefully disable all automation, disable certain features, or fail to maintain the system (for example, by failing to accept software updates). Because owners are ostensibly the closest to the vehicle post-sale, courts and juries may find that they share potential liability.

Drivers: Disabling Features, Improper Use

If a human "driver"[106] is not the owner, that driver may share a distinct liability. For example, even if the owner has maintained and enabled all safety features, the driver might disable them. Or the driver may use the automation features improperly. Or the driver might override the automation, contributing to a crash. Indeed, Google/Waymo reports that nearly all of that company's automated-driving crashes have been caused by human error. So in Levels 1 through 4, potential liability of drivers and owners can be distinct.

Manufacturers and Software Creators: Failure to Warn

Some vehicle manufacturers, such as Tesla, are developing their automated-driving technologies themselves; other manufacturers are working with an outside software vendor (Fiat-Chrysler and Google/Waymo, for example, announced a partnership in 2016).[107] So where an accident involves a company that manufactures both the vehicle and its software—as a Chinese man in 2017 claimed that Tesla's Autopilot function caused his son's death,[108] or a motorcyclist suing GM for a California accident[109]—that results in an easier-to-determine defendant: the manufacturer/developer. But for automated-driving partnerships that split hardware and software roles, potential liability for an automated vehicle's design may be split.

Today's software disputes often hinge on language in end-user license agreements (EULAs). For automated-driving disputes, liability may also be tied to what the manufacturers' EULAs claim and disclaim. Do automated-vehicle purchasers own the *entire* vehicle—hardware and software? Or does the software remain the sole property of manufacturers? The likelier scenario is one where manufacturers own the vehicle's software (General Motors and John Deere presently claim ownership), and customer owners merely license it.[110] Despite manufacturers' claims of ownership, the US Copyright Office has allowed customer owners to break software encryption to repair vehicles[111]—potentially indicating that, even absent software *ownership*, consumers might be able to exert some *control*.

The interaction between software and hardware in automated vehicles raises interesting questions. Would a manufacturer's ownership claims over the software require a heightened duty to update, as well as potential liability from failure to warn?

The issue of software ownership also relates to future auto sales. Could manufacturers who claim software ownership prevent consumers from selling automated vehicles—as Google's license might have prohibited sales of Google Glass on eBay?[112] Would a manufacturer's attempted prohibition of automated-vehicle sales conform to copyright's first-sale doctrine, which permits a purchaser of a copyrighted work such as a DVD to sell it? Does it matter that without the automated-driving software, the hardware (the vehicle itself) would be rendered useless?

Data Providers

In some instances, software designers (such as Google) and other data providers might provide data independently, potentially resulting in a liability split. For example, automated-driving software might use a third party's maps. If the automated vehicle follows incorrect map data—if, say, that data reflects the wrong direction on a one-way street—then the map-data provider might be liable, at least in part.

Automated vehicles will likely be programmed to follow the laws for any given GPS position, and they might use third-party data to determine applicable laws. This is a complicated proposition, since any location (as determined by GPS) is usually governed by myriad, often-overlapping laws and regulations: local and municipal laws, county laws, state laws, federal statutes, federal regulations, and potentially others. If data providers get any one of those legal concentric circles wrong, or fail to timely reflect changes to legislation or caselaw, then the data providers could bear some liability.

Potential Claims

During the transition to a driverless future, the legal standards associated with auto crashes—which have remained largely stable for decades—will likely evolve more quickly. As automated driving becomes more prevalent, improved safety may decrease bodily-injury claims, while sensor damage and AI failures will likely increase property-damage claims and product-liability claims.

Lawsuits about automated driving might involve several types of traditional claims: negligence, product-

liability claims, failure to warn, breach of warranty, and foreseeability. But injured plaintiffs' claims will likely differ depending on their personal standing. Automated vehicles' owners, "drivers," passengers, and pedestrians will likely have distinct claims.

A 50-state survey of negligence, property, and product-liability claims would fill a treatise and is far beyond this article's scope. As such, the analyses below will focus on one state, Minnesota, and how its existing laws could be adapted and applied to automated driving. Because those areas of law tend to closely align across jurisdictions, this state's analysis might indicate potential applications in other jurisdictions.

Negligence

Negligence claims in product-liability cases often hinge on whether a defendant used reasonable care to design products so they are safely used in reasonably foreseeable ways. For example, Minnesota law requires the following elements of negligence: (1) the existence of a duty of care, (2) breach of that duty, (3) injury, and (4) proximate cause.[113] For manufacturers, "a supplier has a duty to warn end users of a dangerous product if it is reasonably foreseeable that an injury could occur in its use."[114]

If an automated vehicle crashes on a wet road, plaintiffs will likely argue that manufacturers either foresaw or should have reasonably foreseen that automated vehicles would navigate wet roads, and the vehicles' design caused injury. But in such a case, one salient question might be this: To what extent was the manufacturer's design of the automated-driving system "reasonable"? When encountering wet roads, should the system have been designed to automatically reduce vehicle speed? Prevent driving altogether?

Strict Liability

Even if a manufacturer is not negligent, having exercised all requisite care, it might still ship vehicles with unsafe defects that trigger strict liability, regardless of manufacturer negligence. Strict liability cases generally fall into three categories: (1) manufacturing defects, (2) design defects, and (3) failure to warn. The Second Restatement of Torts reflects that manufacturers are liable for "unreasonably dangerous" defects even if they "exercised all possible care" to prepare and sell the product, and even without a contractual or purchasing relationship with the user.[115] For automated-driving cases, the vehicles' "users" (including owners, passengers, and potentially third parties) might make similar claims. The Third Restatement slightly shifts liability to failure to warn of "foreseeable risks"—and "foreseeability" in the fast-moving area of automated driving will likely be a moving target.

Manufacturing Defects. In automated vehicles, manufacturing defects might involve either hardware or software. To prove a manufacturing defect under Minnesota law, plaintiffs must establish: (1) the product's defective condition was unreasonably dangerous for its intended use, (2) the defect's existence when the product left the defendant's control, and (3) proximate cause.[116] Hardware manufacturing defects are more familiar to product-liability attorneys today, but in a world where algorithmic tweaks can make the difference between life and death, software manufacturing defects may play a larger role. What if a beta algorithm ships with a flaw that interprets a white semi-trailer as the sky? Or a typo-induced bug in the governing law (25 mph vs. 255 mph)? Are those manufacturing defects or design defects?

Design Defects. A more common claim might be that an automated vehicle has a defective design. In Minnesota, a manufacturer must take "reasonable care" when designing a product to "avoid any unreasonable risk of harm to anyone who is likely to be exposed to the danger when the product is used in the manner for which the product was intended, as well as an unintended yet reasonably foreseeable use."[117] Courts understand that the "reasonable care" analysis balances likelihood and gravity of harm with the burden of effective precaution.[118]

Design, Security, and Safety Updates. Because automated-driving algorithms are improving daily, plaintiffs might argue that yesterday's state-of-the-art algorithm is today's defective design. If today's state-of-the-art algorithms can make more subtle distinctions, but a manufacturer does not push the algorithmic improvements to a 10–year-old car's system, is that a design "defect"? What is the burden of pushing the updated algorithm to an obsolete system? Does it matter that manufacturers could push improvements quickly and inexpensively through over-the-air updates?

Even today, this question is not hypothetical. In September 2016, Chinese researchers discovered security vulnerabilities in Tesla vehicles, permitting unauthorized remote activation of moving vehicles' brakes. Tesla reportedly pushed its fix to vehicles within 10 days.[119]

Sunsetting and Planned Obsolescence. Consumer technology manufacturers face a similar situation with software and operating system (OS) updates. As Microsoft, Apple, and Google improve their OSes, they "sunset" the operating systems, often leaving devices inoperable or vulnerable to malicious hacking. Microsoft, for example, stopped supporting Windows XP in 2014, leaving users of that ancient, 12-year-old

OS vulnerable to security attacks. Smartphones and tablets have even shorter supported-OS lifespans. But while consumers might accept that a $300 tablet they bought in 2013 reached its end-of-supported-life in 2017, they will likely demand a higher standard from a $100,000 automated vehicle.[120] Even with today's smartphones, the FCC has pressured devicemakers to provide details about the frequency and timeliness of their security updates.[121] Manufacturers, regulators, and the courts will likely make similar determinations for automated vehicles. What security-update and algorithm-update frequency is "reasonable"? And can manufacturers permissibly choose to "brick" obsolete vehicles? Or could they allow continued use "at your own risk"?

Failure to Warn. Although automated vehicles promise great safety improvements, plaintiffs will surely argue that they remain dangerous products that require warnings. In Minnesota, suppliers have a duty to warn end users of a dangerous product if "it is reasonably foreseeable that an injury could occur in its use."[122] If that duty is triggered, the supplier has two duties:

1. The duty to give adequate instructions for safe use.

2. The duty to warn of dangers inherent in improper usage.[123]

Legally adequate warnings should:

1. Attract the attention of those that the product could harm;

2. Explain the mechanism and mode of injury; and

3. Provide instructions on ways to safely use the product to avoid injury.[124]

When determining whether the duty to warn exists, the "linchpin" is foreseeability.[125] Minnesota courts analyze an allegedly negligent act, as well as the event causing the damage, by determining the following:

1. **Court determines *no* duty:** Connection is too remote to impose liability as a matter of public policy;

2. **Court determines duty exists:** Consequence was direct, the type of occurrence was or should have been reasonably foreseeable;

3. **Jury considerations:** Adequacy of the warning, breach of duty, causation.[126]

For Minnesota cases involving automated driving Levels 1–4, parties will likely cite the Minnesota Supreme Court's decision in *Glorvigen v. Cirrus Design Corp.*, which involved an airplane's autopilot mode and the manufacturer's alleged failure to warn.[127] That court held that providing the pilot with written instructions was sufficient: "[T]here is no duty for suppliers or manufacturers to *train* users in the safe use of their product."[128]

Automated-driving vehicles will likely raise distinct legal questions. For partial automation, where NHTSA Level 2 requires operator intervention on "short notice," what type of warning sufficiently defines "short"? May the manufacturers engage in a cost-benefit analysis: Opting not to spend millions of dollars to provide faster notice, where any user performance improvement would be negligible?

Full automation such as Level 5 will raise even more-interesting questions. In contrast to "autopiloted" airplanes, which still require training, expertise, and *shared control*, what standard would apply to fully autonomous vehicles, which might require ***no*** training? What type of written warning might be sufficient? Given automated driving's potential benefits to the disabled, must manufacturers provide warning sufficient to accommodate "drivers" of fully automated vehicles who are deaf, blind, or both?

Misrepresentation

Injured plaintiffs will likely argue that automated-vehicle manufacturers' advertising and statements reflect fraudulent or negligent misrepresentations. For example, for human-machine cooperation in Levels 1–4, to what extent can a manufacturer permissibly assert that human control is "rare"? What if the manufacturer cites testing data—regarding automation/human handoffs, crash data, injuries, or other parameters—and real-world data varies dramatically? Can a manufacturer tout the benefits of reading email or watching movies? (In a 2016 fatal crash involving Tesla's Autopilot, the driver was reportedly watching a Harry Potter movie.[129]) What evidentiary support must manufacturers compile to support claims that their automation is "safe"?

Breach of Warranty

Plaintiffs will also likely argue that automated-vehicle manufacturers have breached their warranties of quality that were created through sales and marketing. The UCC's applicable provisions include those on express warranties and implied warranties.[130]

Express warranties. What if a manufacturer expressly advertises that its fully autonomous vehicle is "as safe as human drivers"? Under UCC § 2–313, does the

statement set some standard of warranted performance? As safe as an average driver? The best drivers? Distracted drivers? The worst? Or is the claim of "safety" so ambiguous that it's mere advertising puffery? What if that statement comes not from a manufacturer, but from a dealer who accompanies an oral statement with a written agreement's merger clause?[131]

Implied warranties. Under UCC §§ 2-314(2)(a)–2-314(2)(f), what are potential implied warranties? In this new product category, what attributes make automated vehicles "fit for the ordinary purpose for which such goods are used"? And what of other categories that require products to "conform to the promise or affirmations of fact" and "pass without objection in the trade"?

Third-party beneficiaries? Who beyond the immediate buyer does the warranty benefit? Many jurisdictions will extend warranties to third-party beneficiaries, potentially including family, household members, guests, or any other person, with different jurisdictions providing varying levels of protection.

Who sold to whom? Another consideration is privity. What if buyers purchase vehicles directly from manufacturers? Tesla, for example, eschews dealerships for "stores," establishing a direct relationship with buyers. In those direct purchases, third-party and pass-through warranties under UCC § 2-318 may be inapplicable. But § 2-318 would likely apply to more traditional dealership models, where warranties would probably pass through to third-party beneficiaries.

Disclaimers beware. What if the manufacturer includes a disclaimer? Indeed, most implied warranties can be disclaimed unless barred by state statute or by the Magnuson Moss Warranty Act. Express warranties cannot be disclaimed, but manufacturers can include well-drafted merger clauses to limit potential liability to written warranties. As such, manufacturers' and dealers' disclaimers could well extinguish many potential warranty claims.

Global No-Fault Compensation Act?

To address the thorny liability issues, policy think tank RAND has suggested a no-fault insurance system that is similar to the 1986 National Childhood Vaccine Injury Act (a no-fault system to compensate vaccine recipients with serious adverse reactions). Just as the vaccine act, which sought to reduce the possibility of lawsuit-besieged manufacturers scaling back vaccine production, a similar no-fault auto-insurance act could strike a similar balance. Such a policy might encourage the development of life-saving technology, while minimizing market forces that might encourage technological stagnation.

Insurance Implications: Autonomy Discounts? Penalize Driving Manually?

Few businesses have a greater interest in regulating automated driving than insurers. Large insurance groups have opined that, while traditional underwriting criteria (*e.g.,* driver's number of accidents, miles driven, parking location) will probably still apply, automated driving might lead to greater emphasis on the car's make, model, and style. [132] Also important will be the use of telematics devices (*i.e.,* "black boxes") that monitor driver behavior, leading to potential premium discounts and increases.

Currently, the insurance industry offers discounts for drivers who use black boxes, but adoption has been lower than expected—likely because of the privacy implications. However, the National Association of Insurance Commissioners believes that in the next five years, usage of black boxes will increase to 20 percent of drivers.[133] If the automated-driving industry's forecasts are accurate, automated driving's prevalence will explode. If automated vehicles achieve higher safety than today's vehicles, insurers may well provide discounts for automation. Conversely, Tesla CEO Elon Musk has opined that manual driving, which he views as objectively unsafe, might eventually be outlawed.[134] If that vision proves true, we may see a world where only the wealthy can drive, and grandchildren might beg grandparents, "Tell me about when you used to drive yourself!"

Criminal Implications: Algorithmic Law Enforcement?

Observance of today's traffic laws is encouraged (through fines and potential imprisonment). Today's enforcement of driving laws depends on several factors: law enforcement (1) observing the behavior, (2) determining that a law was broken, and (3) gathering evidence. As such, today's enforcement has limited success, if gauged by the percentage of drivers who abide by posted speed limits. (How many cars on a 55-mph highway drive under 55 mph?)

But with automated vehicles, today's sporadically enforced laws might be enforced universally, automatically, and algorithmically. With automated systems, any potential "recklessness" can be virtually eliminated by algorithm. Self-driving cars could be programmed to obey the law. Automated vehicles could recognize laws of overlapping jurisdictions (city, county, state, federal), import traffic rules (*e.g.,* speed limits, turn on red), and implement those laws through the automated-driving

system. In a fully automated (Level 5) future, the number of traffic violations could theoretically be reduced to nearly zero.

Legal Code's Enforcement Through Computer Code

Of course, the systems could also be programmed to recognize legal exceptions currently permitted by human judgment. For example, ambulances would likely be permitted to speed, and vehicles might be permitted to cross a double center line to avoid stalled cars or downed trees.

But in partially automated Levels 1–4, where the human and computer co-operate the vehicle, to what extent will (or should) manufacturers be required to implement driving laws? For example, should a Level 4 or Level 5 system permit a human driver to speed—while taking an injured child to the emergency room? Avoiding an attacker? Making a flight? Meeting a client?

In addition, if humans override the laws and systems' algorithms, will those deviations be logged, as flags to review potential legal violations? Should manufacturers be compelled to compile and produce infractions to requesting governmental entities? Of course, these questions raise significant privacy implications.

No doubt, implementing legal code through algorithmic code is tempting and could well bring significant benefits. But if regulators choose to implement legal codes through computer code, the questions about potential effects of those programmed laws—and their permitted exceptions and privacy implications—will be many.

What Will Happen to Pretextual Stops?

Today, many criminal charges stem from police stopping a vehicle for an alleged infraction (*e.g.,* broken tail light) and then discovering evidence of a more serious crime (*e.g.,* drugs, parole violation, felony gun possession). In 1996, a unanimous Supreme Court held that traffic stops do not violate the Fourth Amendment "even if a reasonable officer would not have stopped the motorist absent some additional law enforcement objective."[135] But legal technologist and former public defender David Collarusso has correctly questioned whether—in a world where automated vehicles obey every traffic law—pretextual stops will be rendered extinct.[136] Because automated vehicles' routes will likely be logged, will courts instead permit police to use data analytics, based on driving patterns from "flight plans," to justify an automated vehicle's detention? "You probably realize that I pulled you over because you made nine five–minute stops in a neighborhood with statistically high rate of drug crimes."

Privacy Implications: Why Did My Car Drive Me to McDonalds?[137]

The promise of automated driving also generates potential privacy concerns. Automation's benefits include a vehicle's ability to take users on the most-efficient routes, as well as the vehicles' communication with other vehicles (*e.g.*,V2V) to both expedite trips and increase safety. But those activities all involve data, and that data could be tracked. As such, automated driving will implicate several privacy concerns.

If automated vehicles will log trip routes (as Google Maps and Apple Maps currently log users' routes, depending on privacy settings), and if those routes are readily shared with manufacturers, will riders still have a "reasonable expectation of privacy"? Would a governmental subpoena of Google's location logs differ from the warrantless GPS tracking of a vehicle, which the Supreme Court in 2012 held was unconstitutional?[138] The following year, police officers used the Stored Communications Act to request of mobile-phone carriers' location data for three suspects, without a warrant or probable cause, and the Fifth Circuit held that the request was not a *per se* violation of the Fourth Amendment.[139] The Seventh Circuit has held similarly.[140]

Would the warrantless subpoena of automated-driving logs be different? The Supreme Court has held that smartphones are "a digital record of nearly every aspect of [Americans'] lives—from the mundane to the intimate," but would the same hold true for a utilitarian, single-purpose vehicle? Also, as the public becomes increasingly aware of the ability of software companies and governmental entities to track them, does the "reasonable expectation of privacy" change over time, on a sliding scale?

Advertising may also play a role in any privacy debate. A company that has invested heavily in automated driving is Google, which derives significant revenue from advertising. Any advertising company's business plan might include leveraging automated driving by advertising nearby services. Could an advertising company permissibly extend advertising benefits to include unplanned stops at advertisers?[141] Encourage fuel-ups at advertiser stations? Track users' trips to sensitive areas (*e.g.*, psychiatrist, mosque, gay bar, abortion clinic)? To some extent, the ability to track is already present in today's smartphones. But automated driving—and the ability to physically change users' locations—may heighten any perceived privacy implications.

Ethical Implications

As automated vehicles make decisions currently made by humans, machines will be more likely to

know—before the decisions are made—when those automated decisions will increase the probability of injury or death.

Algorithms as God: Who Lives and Who Dies?

Consider this scenario:[142] A single-passenger fully automated car on an icy mountain road encounters 10 pedestrians. The car calculates two options:

Option 1: Stay on the icy road, save its passenger, and kill the 10 pedestrians.

Option 2: Swerve off the mountain road, kill its passenger, but save the 10 pedestrians.

Option 2 is the most utilitarian: It results in one death, not 10. But the car's driver and purchaser would obviously prefer Option 1, which kills more people but saves the purchaser. In this way, algorithms can and will determine ethical quandaries beforehand—something the law has considered for human liability (*e.g.*, premeditation), but not yet with machines.

If a manufacturer chooses Option 1, might lawyers representing the 10 dead pedestrians claim that the fault lies with:

1. The owner's enabling Option 1 (if it was an option)?
2. The vehicle manufacturer's choice of implementing Option 1—as default, or even as an option?
3. The software developer's even considering Option 1 as a factor in the first place?

Even if the manufacturer chooses to implement Option 2, lawyers will likely make similar arguments. Of course, one could argue "don't program any algorithm—omit Option 1 and Option 2." But since the vehicle would continue on the road, killing the pedestrians, that non-choice is really a choice (Option 1).[143]

Professor Smith has opined that under either scenario, Option 1 or Option 2, plaintiffs will likely sue, and, under both scenarios, they will likely succeed. It's easy to understand the view that for automated-driver manufacturers, liability will be an inevitability.

As such, it's not difficult to consider that manufacturers may take that liability into account when creating algorithms. Besides calculating the number of injuries or deaths (utilitarianism), one can easily imagine the algorithm considering other potential factors:[144]

- Save the youngest (*i.e.*, children)?
- Save the middle-aged (*i.e.*, those the most earning power)?
- Save the healthiest? (Or forego the best organ donors?)
- Save advertisers?
- Save US citizens?

These questions are, no doubt, uncomfortable and perhaps even ghoulish. So perhaps a third solution is this:

Option 3: If the system detects a low threshold of safety (*e.g.*, snow, rain, blind corners), the vehicle either slows or parks, remaining disabled until conditions improve.

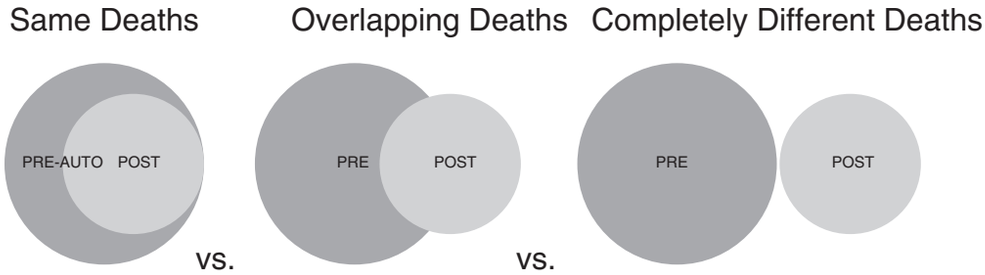
Of course, choosing Option 3 would allow manufacturers to avoid the Sophie's Choice of Options 1 or 2. But query whether busy commuters would accept their inability to travel because their immobile car displays the error: "Too snowy."

With our automated-driving future careening toward us, the question of what factors the algorithms should consider (and those they should not) will eventually need answering by software designers, manufacturers, regulators, legislators, or courts.

Focus on Utilitarian Good or on the Means to the End?

Beyond an individual case, as discussed above, experts have had fascinating discussions[145] about the murkiness of ethics on autonomous vehicles generally, including their societal effect. For example, if the current death rate from vehicles is 32,000 deaths per year, and under autonomous vehicles, that net rate is 16,000 deaths, then society would have "saved" 16,000 lives—which is good, right? But that is just the *net* rate. What if all 16,000 of the new deaths were computer-caused and wouldn't have happened pre-autonomy? So if autonomy saves 16,000 lives, but the algorithms choose—Skynet style—those who live and those who die, is that okay? Or does society (as reflected in our laws and regulations) pre-

Exhibit 5—Difficult Questions Posed by Autonomous Vehicles



fer our pre-automation randomness and 16,000 more deaths? These are not easy questions, as illustrated in Exhibit 5 by five potential scenarios, increasing in ethical difficulty from left to right.

Bring It Home

Automated driving already is here and its development is progressing steadily. Gauging by NHTSA's enthusiasm for the potential safety benefits of automated driving, as well as manufacturers' space race to deliver, automated driving may arrive sooner than many of us expect. If so, the question of how our laws and regulations should apply or adjust to automated driving will be firmly within the province of our legislators, regulators, insurers, litigators, and judges. For certain, this will be a wild ride.

Notes

1. *See https://www.tesla.com/presskit/autopilot.*
2. *See https://venturebeat.com/2017/06/04/self-driving-car-timeline-for-11-top-automakers/.*
3. *See https://www.law.upenn.edu/fac/pwagner/law619/f2001/week15/easterbrook.pdf.*
4. This emerging technology has attracted several names: "autonomous vehicles," "driverless cars," and "self-driving cars" are a few. Because this article discusses the range of capabilities—from manual vehicles to partial assistance to full autonomy—it will follow the Stanford Center for Internet and Society's convention by using the term "automated driving."
5. *See https://www.hsdl.org/?view&did=795644.*
6. *See http://www.nhtsa.gov/staticfiles/rulemaking/pdf/Autonomous-Vehicles-Policy-Update-2016.pdf.*
7. *See https://www.engadget.com/2016/01/30/tesla-hires-chip-veteran-for-self-driving-tech/.*
8. *See https://www.wired.com/2016/01/gm-and-lyft-are-building-a-network-of-self-driving-cars/.*
9. *See http://video.cnbc.com/gallery/?video=3000505167.*
10. *See http://orfe.princeton.edu/~alaink/SmartDrivingCars/PDFs/IHS%20_EmergingTechnologies_AutonomousCars.pdf.*
11. *See https://www.tesla.com/presskit/autopilot* ("Drivers must keep their hands on the steering wheel).
12. *See http://www.autonews.com/article/20160112/OEM04/160119857/gm-delays-super-cruise-rollout.*
13. *See http://www.toyota.com/safety-sense/.*
14. *See https://waymo.com/ontheroad/.*
15. *See http://newsroom.aaa.com/2018/01/americans-willing-ride-fully-self-driving-cars/; https://www.consumerreports.org/autonomous-driving/consumer-confidence-in-self-driving-technology-is-increasing-aaa-study-finds/.*
16. Before that, Google claims that the prior 17 crashes involving their self-driving cars were all caused by human error, either by Google's drivers or by other cars, not the automation. *See http://www.theverge.com/2016/2/29/11134344/google-self-driving-car-crash-report.*
17. *See https://www.tesla.com/blog/tragic-loss.*
18. *See https://www.tesla.com/blog/tragic-loss.*
19. *See https://www.ntsb.gov/news/press-releases/Pages/PR20170912.aspx.*
20. *See http://blog.caranddriver.com/fatal-tesla-crash-wont-slow-federal-push-for-autonomous-cars/.*
21. *See http://www.businessinsider.com/report-10-million-self-driving-cars-will-be-on-the-road-by-2020-2015-5-6.*
22. *See http://orfe.princeton.edu/~alaink/SmartDrivingCars/PDFs/IHS%20_EmergingTechnologies_AutonomousCars.pdf.*
23. *See https://www.ieee.org/about/news/2014/14_july_2014.html.*
24. HLDI (2049) and HIS (2050).
25. *See https://www.sae.org/works/committeeHome.do?comtID=TEVAVS.* Society of Automotive Engineers.
26. *See http://cyberlaw.stanford.edu/blog/2013/12/sae-levels-driving-automation.*
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